Question and Answer:

1. Find Fourier transform of $f(t) = \begin{cases} \sin t & for\ \ 0<t<\pi \\ 0 & otherwise \end{cases}$

Solution: $F(\omega)=\int_{-\infty}^{\infty} e^{-i\omega t} f(t)dt = \int_{0}^{\pi} e^{-i\omega t}\sin t dt$

$$=\left[\frac{e^{-i\omega t}}{i^2\omega^2+1}\{-i\omega\sin t-\cos t\}\right]_0^{\pi}$$

$$=\frac{-1}{1-\omega^2}\left[e^{-i\omega t}\{i\omega\sin t+\cos t\}\right]_0^{\pi}$$

$$=\frac{-1}{1-\omega^2}\left[e^{-i\omega\pi}\{i\omega\sin\pi+\cos\pi\}-\cos 0\right]$$

$$=\frac{-1}{1-\omega^2}\left[-e^{-i\omega\pi}-1\right]$$

$$\boxed{F(\omega)=\frac{1}{1-\omega^2}\left[e^{-i\omega\pi}+1\right]}$$

2. Find the Fourier transform of the function,

$$f(t) = \begin{cases} a^2-t^2 & for\ \ |t|\le a \\ 0 & for\ \ |t|>a \end{cases}$$

Solution: $F(\omega)=\int_{-\infty}^{\infty} e^{-i\omega t} f(t)dt = \int_{-a}^{a}(a^2-t^2)e^{-i\omega t}dt$

$$=\left\{(a^2-t^2)\left[\frac{e^{-i\omega t}}{-i\omega}\right]-(2t)\left[\frac{e^{-i\omega t}}{(-i\omega)^2}\right]+2\left[\frac{e^{-i\omega t}}{(-i\omega)^3}\right]\right\}_{-a}^{a}$$

$$=\left\{(a^2-t^2)\left[\frac{e^{-i\omega t}}{-i\omega}\right]\right\}_{-a}^{a}-\left\{(2t)\left[\frac{e^{-i\omega t}}{(-i\omega)^2}\right]\right\}_{-a}^{a}+2\left[\frac{e^{-i\omega t}}{(-i\omega)^3}\right]_{-a}^{a}$$

$$=\left\{(a^2-t^2)\left[\frac{e^{-i\omega t}}{-i\omega}\right]\right\}_{-a}^{a}-\left\{(2t)\left[\frac{e^{-i\omega t}}{(-i\omega)^2}\right]\right\}_{-a}^{a}+2\left[\frac{e^{-i\omega t}}{(-i\omega)^3}\right]_{-a}^{a}$$

$$=0-\left[2a\left[\frac{e^{-i\omega a}}{(-i\omega)^2}\right]+2a\left[\frac{e^{i\omega a}}{(-i\omega)^2}\right]\right]+2\left[\frac{e^{-i\omega a}}{(-i\omega)^3}-\frac{e^{i\omega a}}{(-i\omega)^3}\right]$$

$$=\frac{-2a}{\omega^2}\left[e^{-i\omega a}+e^{i\omega a}\right]-\frac{2}{i\omega^3}\left[e^{-i\omega a}-e^{i\omega a}\right]$$

$$= \frac{-4a}{\omega^2}\frac{\left[e^{-i\omega a}+e^{i\omega a}\right]}{2}+\frac{4}{\omega^3}\left[\frac{e^{i\omega a}-e^{-i\omega a}}{2i}\right]$$

$$= \frac{-4a}{\omega^2}\cos\omega a+\frac{4}{\omega^3}\sin\omega a=\frac{4}{\omega^3}\left[\sin\omega a-\omega a\cos\omega a\right]$$

$$\boxed{F(\omega)=\frac{4}{\omega^3}\left[\sin\omega a-\omega a\cos\omega a\right]}$$

3. Find the Fourier sine transform of $f(t) = \dfrac{1}{t(1+t^2)}, t>0.$

Solution: $F_s(\omega)=\int_0^\infty f(t)\sin\omega t\,dt$ -----------------(1)

$$=\int_0^\infty \frac{1}{t(1+t^2)}\sin\omega t\,dt$$

$$F_s'(\omega)=\int_0^\infty \frac{t}{t(1+t^2)}\cos\omega t\,dt$$

$$=\int_0^\infty \frac{1}{(1+t^2)}\cos\omega t\,dt \text{ ----------------(2)}$$

$$F_s''(\omega)=\int_0^\infty \frac{-t}{(1+t^2)}\sin\omega t\,dt=-\int_0^\infty \frac{t^2}{t(1+t^2)}\sin\omega t\,dt$$

$$=-\int_0^\infty \frac{(t^2+1-1)}{t(1+t^2)}\sin\omega t\,dt=-\int_0^\infty \frac{\sin\omega t}{t}dt+\int_0^\infty \frac{1}{t(1+t^2)}\sin\omega t\,dt$$

$$=-\int_0^\infty \frac{\sin\omega t}{t}dt+F_s(\omega)==-\frac{\pi}{2}+F_s(\omega)$$

$$\Rightarrow F_s''(\omega)-F_s(\omega)=-\frac{\pi}{2}\Rightarrow (D^2-1)F_s(\omega)=-\frac{\pi}{2}$$

$$\Rightarrow F_s''(\omega)-F_s(\omega)=-\frac{\pi}{2}$$

$$\Rightarrow F_s(\omega)=C_1e^{\omega}+C_2e^{-\omega}+\frac{\pi}{2} \text{ ------------(3)}$$

$$\Rightarrow F_s'(\omega)=C_1e^{\omega}-C_2e^{-\omega} \text{ ------------(4)}$$

Put $\omega=0$ in (3), then we get, $C_1+C_2=0$ (using (1))

Put $\omega=0$ in (4), then we get, $C_1-C_2=\int_0^\infty \frac{dt}{1+t^2}$ (using (2)) i.e., $C_1 - C_2 = \frac{\pi}{2}$; on solving, we get, $C_1 = 0$ *and* $C_2 = -\frac{\pi}{2}$

Substituting in (3), we get, $F_s(\omega) = \sqrt{\frac{\pi}{2}}\left(1 - e^{-\omega}\right)$

4. Find $f(t)$ satisfying the integral equation, $\int_0^\infty f(t)\cos\omega t dt = e^{-\omega}$

Solution: We know that, $F_c(\omega) = \int_0^\infty f(t)\cos\omega t dt = e^{-\omega}$ (given)

Therefore, $f(t) = \frac{2}{\pi}\int_0^\infty F_c(\omega)\cos\omega t d\omega = \frac{2}{\pi}\int_0^\infty e^{-\omega}\cos\omega t d\omega$

$$= \frac{2}{\pi}\left(\frac{e^{-\omega}}{\left(\omega^2+1\right)}\left[-\cos\omega t + \omega\sin\omega t\right]\right)_0^\infty = \frac{2}{\pi\left(\omega^2+1\right)}\left(e^{-\omega}\left[-\cos\omega t + \omega\sin\omega t\right]\right)_0^\infty$$

$$= \frac{2}{\pi\left(\omega^2+1\right)}\left(e^{-\omega}\left[-\cos\omega t + \omega\sin\omega t\right]\right)_0^\infty = \frac{2}{\pi\left(\omega^2+1\right)}\left(0 - [-1]\right) = \frac{2}{\pi\left(\omega^2+1\right)}$$

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